Amanda Mai

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EDUCATION

University of California, Los Angeles (UCLA) **Los Angeles, CA**

Bachelor of Science in Computer Engineering **Sept. 2022 – June 2026**

Relevant Coursework:

OS Principles, Software Construction, Algorithms and Complexity, Systems and Signals, Digital Design Laboratory, Electrical and Electronic Circuits, Digital Electronic Circuits, Computer Systems Architecture, Systems Design Capstone

WORK EXPERIENCE

TrueCx SWE Intern | Jun. 2025 – Sept. 2025

- Redesigned AI scoring system for consistent, accurate responses using Gemini API, prompt engineering, and Postgres
- Developed comprehensive QA test matrices and cases, ensuring thorough bug tracking and documentation in Jira.
- Automated system monitoring via Python scripts to track API health and pipeline performance.
- Prototyped and validated RESTful APIs using Postman to ensure correctness and performance of data pipelines.
- Established and maintained Confluence documentation spaces to support team knowledge sharing and onboarding

PROJECT EXPERIENCE

sEMG Typing Decoder (emg2qwerty) | Jan. 2026 – Mar. 2026

Designed a deep learning pipeline in PyTorch to decode typed characters from surface electromyography (sEMG) signals using the emg2qwerty dataset.

- Implemented and evaluated CNN, GRU, LSTM, and hybrid CNN–GRU architectures for sequential signal decoding.
- Programmed preprocessing and augmentation techniques including Z-score normalization, Gaussian noise injection, time warping, and amplitude scaling to improve model robustness.
- Optimized the LSTM architecture, achieving a 14.8% character error rate (CER) on test data.

Out-of-Order Processor Simulator | Sept. 2025 – Dec. 2025

Designed a cycle-accurate out-of-order processor simulator in C++, modeling fetch, dispatch, scheduling, execution, and retirement stages with configurable functional units, result buses, and fetch width.

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- programmed reservation stations, register renaming readiness tracking, and in-order retirement to model data hazards.

Verilog Obstacle Game | Nov. 2024 – Dec. 2024

Developed a Verilog game on the Basys 3 FPGA with character movement, obstacle avoidance, and dynamic difficulty modes.

- Programmed obstacle generation and movement logic, using 7-segment LEDs to visually represent the game state
- Integrated button controls to enable user interaction, allowing the player to control the character's actions, restart the game, and display the score upon game over, all within a real-time embedded environment.

Study Group Finder Website | Jan. 2024 – Mar. 2024

Worked with a team to develop a study group matching app that functions similarly to a dating app

- Implemented a form that validated user’s input and wrote CRUD operations to store encrypted information in MongoDB
- Designed accessible and user friendly UI components by incorporating readable buttons, textboxes, and scaling

ADDITIONAL SKILLS

Technical: C++, Python, CAD, Java, Javascript, LTSpice simulation, Verilog, Postman, Wireshark, Git, Matlab, Latex

Certifications: Seal of Biliteracy, fluent in written and spoken Vietnamese